



**Assignment\_02**  
**Sec\_01\_02\_03**  
**Summer\_2024**

**Remark:** Release date: 5.7.2024

Submission date: 11.07.2024 (6 pm) (online submission because of shortage of time)

**Total mark is 45. It will be converted to 20 later**

1. Let  $V = \left\{ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}$  and  $W = \{ A \in V : A^2 = A \}$ . Is  $W$  a subspace of  $V$ ? [5]

2. Let  $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$  and consider the following subset  $V$  of the 2-dimensional vector space  $\mathbb{R}^2$ , [10]

$$\{x \in \mathbb{R}^2 \mid Ax = 5x\}.$$

- (i) Prove that the subset  $V$  is a subspace of  $\mathbb{R}^2$
- (ii) Find a basis for  $V$  and determine the dimension of  $V$

3. Find the basis for the row and column spaces of the following matrix: [15]

$$A = \begin{pmatrix} 1 & 4 & 5 & 6 & 9 \\ 3 & -2 & 1 & 4 & -1 \\ -1 & 0 & -1 & -2 & -1 \\ 2 & 3 & 5 & 7 & 8 \end{pmatrix}$$

4

Determine the rank of the matrix  $A$ :

[10]

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

5. **Vector Space:** A set  $V$  equipped with two binary operations addition and scalar multiplication is called a vector space over the field  $F$ , if  $V$  satisfies the following 10 axioms, [5]

- |                                                                              |                                                                  |
|------------------------------------------------------------------------------|------------------------------------------------------------------|
| (i) $u + v \in V$ for all $u, v \in V$                                       | (vi) $ku \in V$ for all $u \in V$ and $k \in F$                  |
| (ii) $u + v = v + u$                                                         | (vii) $a(u + v) = au + av$ for all $a \in F$ and $u, v \in V$    |
| (iii) $u + (v + w) = (u + v) + w$                                            | (viii) $(a + b)u = au + bu$ for all $a, b \in F$ and $u \in V$ . |
| (iv) There exists a $0 \in V$ s. t. $u + 0 = u$ for all $u \in V$            | (ix) $(ab)u = a(bu)$ for all $a, b \in F$ and $u \in V$          |
| (v) There exists a $-u \in V$ for all $u \in V$ , such that $u + (-u) = 0$ . | (x) $1u = u$ , where $1 \in F$ and for all $u \in V$ .           |

(a) Suppose  $u = (u_1, u_2)$  the multiplication of  $cu$  is defined to produce  $(cu_1, 0)$  instead of  $(cu_1, cu_2)$ . With usual addition in  $\mathbb{R}^2$ , which of the eight conditions are not satisfied.?